ENEE 323: Digital Signal Processing

Project: Fourth-Order IIR Filter Analysis

March 24, 2026

Name: Yahya Mouamil UID: 119873526

Setup

Problem 0

Consider the fourth-order IIR filter

$$6y[n]-9y[n-1]+9y[n-2]+2y[n-3]-2y[n-4] = 6x[n]+4x[n-1]-4x[n-2]-4x[n-3]-x[n-4].$$

The system is initially at rest.

Solution 0

Taking the z-transform under initial rest gives

$$H(z) = \frac{Y(z)}{X(z)} = \frac{6 + 4z^{-1} - 4z^{-2} - 4z^{-3} - z^{-4}}{6 - 9z^{-1} + 9z^{-2} + 2z^{-3} - 2z^{-4}}.$$

So the numerator and denominator are

$$B(z) = 6+4z^{-1}-4z^{-2}-4z^{-3}-z^{-4}, \qquad A(z) = 6-9z^{-1}+9z^{-2}+2z^{-3}-2z^{-4}.$$

Part (i): Open-Loop Impulse Response

Problem

i

Iterate the difference equation to compute the impulse response samples $h[0 : N_{\text{ceil}}]$, where N_{ceil} is the largest time index n such that

$$|h[n]| \leq 10^{40}.$$

Also plot $h[0 : 99]$ as a bar graph.

Solution

i

For the impulse response, set

$$x[n] = \delta[n].$$

Then the output is $h[n]$. Numerically iterating the filter shows that the response eventually grows very rapidly, so the system is not bounded in impulse response.
The largest index satisfying $|h[n]| \leq 10^{40}$ was found to be

$$\boxed{N_{\text{ceil}} \approx 336.}$$

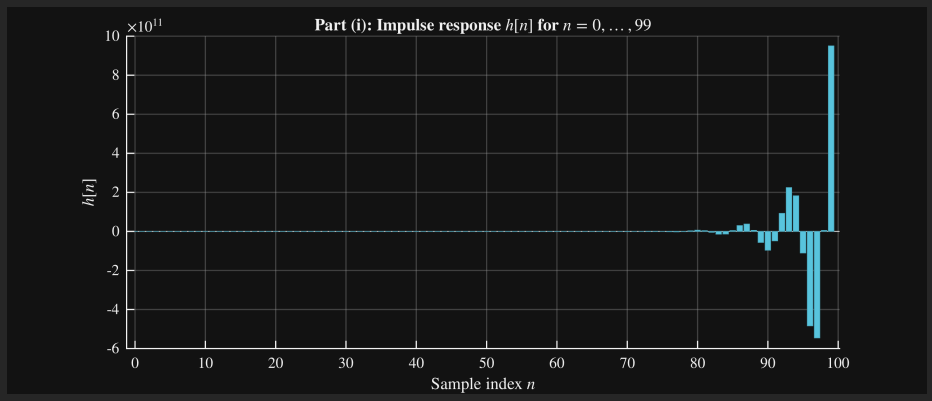

Figure 1: Open-loop impulse response $h[n]$ for $n = 0, \ldots, 99$.

The bar plot already suggests that the response is not decaying to zero. Instead, the magnitude grows, which is a warning sign of instability.

Part (ii): Pole-Zero Plot and Stability

Problem ii

Use the zero-pole plot of $H(z)$ and compute the poles of $H(z)$. Let $R_{\max}$ be the magnitude of the outermost pole. Confirm that

$$\log_{10}(R_{\max}) \approx \frac{40}{N_{\text{ceil}}}.$$

Why is this approximation valid?

Solution ii

The poles are the roots of

$$6z^4 - 9z^3 + 9z^2 + 2z - 2 = 0.$$

Numerically, the largest pole magnitude is

$$\boxed{R_{\max} \approx 1.314446.}$$

Since $R_{\max} > 1$, at least one pole lies outside the unit circle, so the system is **BIBO unstable**.

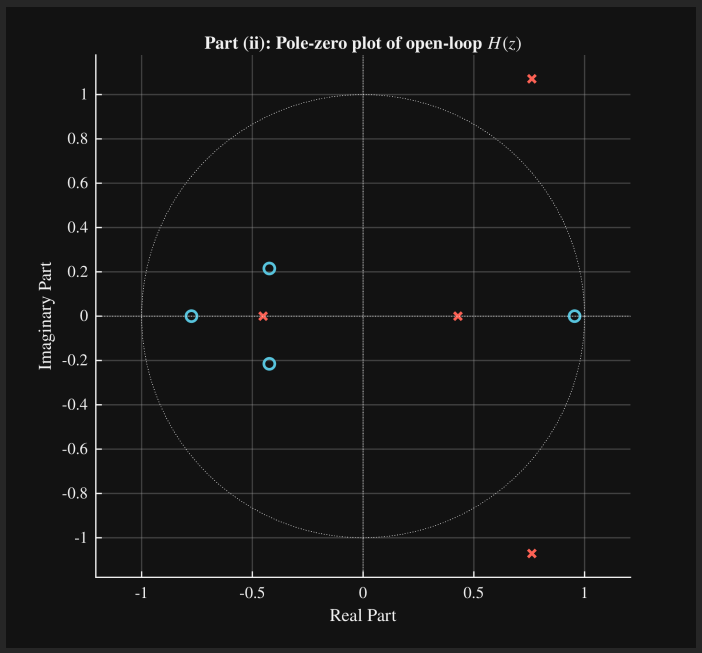

Figure 2: Pole-zero plot of the open-loop transfer function $H(z)$.

Solution ii

Now compare the two quantities:

$$\log_{10}(R_{\max}) \approx \log_{10}(1.314446) \approx 0.118743,$$

and

$$\frac{40}{N_{\text{ceil}}} \approx \frac{40}{336} \approx 0.119048.$$

These are very close, which confirms the approximation.

> This works because for large n, the impulse response is dominated by the pole with largest magnitude. So $h[n]$ behaves roughly like $C(R_{\max})^n$. When $|h[n]|$ reaches about 10^{40}, we have
>
> $$(R_{\max})^{N_{\text{ceil}}} \approx 10^{40}.$$
>
> Taking $\log_{10}$ gives
>
> $$N_{\text{ceil}} \log_{10}(R_{\max}) \approx 40,$$
>
> which leads to
>
> $$\log_{10}(R_{\max}) \approx \frac{40}{N_{\text{ceil}}}.$$

Therefore,

$$\boxed{\text{The open-loop filter is BIBO unstable.}}$$

Part (iii): Closed-Loop System Function

Problem iii

For the feedback configuration with gain $K > 0$, write the closed-loop system function

$$G(z) = \frac{Y(z)}{X(z)}$$

in the standard form $B(z)/A(z)$.

Solution iii

From the diagram, the feedback is **negative feedback**, so the signal entering the filter is

$$E(z) = X(z) - K\,Y(z).$$

Since

$$Y(z) = H(z)E(z),$$

we get

$$Y(z) = H(z)\big(X(z) - K\,Y(z)\big).$$

Rearranging,

$$Y(z)\big(1 + KH(z)\big) = H(z)X(z).$$

Hence

$$G(z) = \frac{Y(z)}{X(z)} = \frac{H(z)}{1 + KH(z)}.$$

Using $H(z) = B(z)/A(z)$,

$$G(z) = \frac{B(z)}{A(z) + KB(z)}.$$

So the closed-loop numerator is still

$$B(z) = 6 + 4z^{-1} - 4z^{-2} - 4z^{-3} - z^{-4},$$

and the closed-loop denominator is

$$A_{\text{cl}}(z) = A(z) + KB(z).$$

Therefore,

$$\boxed{G(z) = \frac{6 + 4z^{-1} - 4z^{-2} - 4z^{-3} - z^{-4}}{(6 + 6K) + (-9 + 4K)z^{-1} + (9 - 4K)z^{-2} + (2 - 4K)z^{-3} + (-2 - K)z^{-4}}}.$$

Part (iv): Find the Smallest Stabilizing Gain

Problem iv

Determine the smallest value of K, accurate to three decimal digits, such that the closed-loop system is BIBO stable.

Solution iv

For BIBO stability, all poles of the closed-loop transfer function must lie strictly inside the unit circle. So the denominator

$$A_{\text{cl}}(z) = A(z) + KB(z)$$

was checked as K varied over the interval $[0, 5]$.
The plot below shows the maximum pole magnitude $R_{\max}$ of the closed-loop system as a function of K.

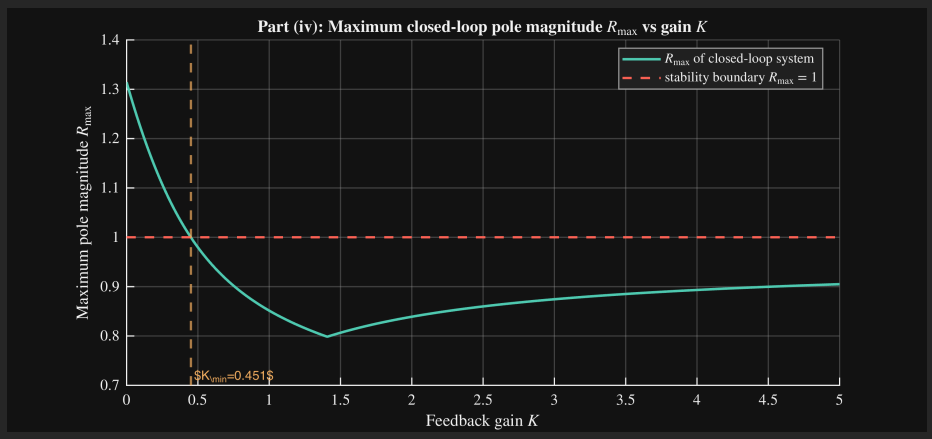

Figure 3: Maximum closed-loop pole magnitude versus gain K. Stability occurs when $R_{\max} < 1$.

The first value of K for which the closed-loop poles move inside the unit circle is

$$\boxed{K_{\min} = 0.451.}$$

Thus, the smallest gain accurate to three decimal places that stabilizes the system is

$$\boxed{K = 0.451.}$$

Part (v): Closed-Loop Impulse Response

Problem v

Using the stabilizing value of K found in part (iv), compute the closed-loop impulse response $g[n]$ for $n = 0 : 999$. Plot it as a continuous graph and confirm that the oscillations decay.

Solution v

Using

$$\boxed{K = 0.451,}$$

the closed-loop system is stable, so its impulse response should decay as $n \to \infty$.
The impulse response $g[n]$ for $n = 0, \ldots, 999$ is shown below.

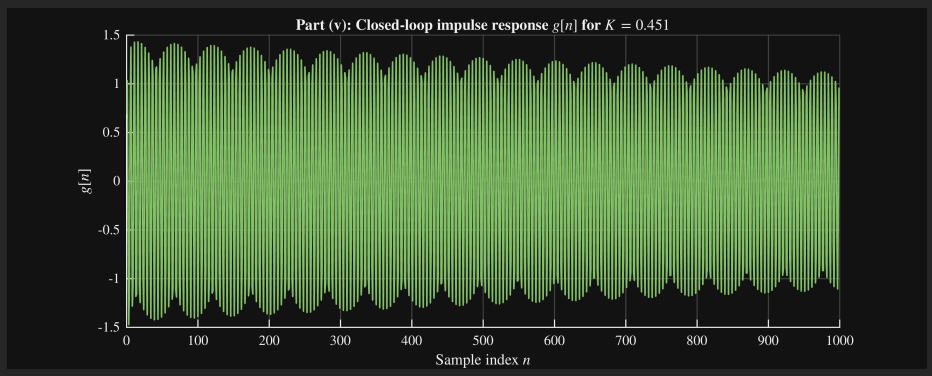

Figure 4: Closed-loop impulse response $g[n]$ for $K = 0.451$.

To make the decay easier to see, the magnitude $|g[n]|$ is also plotted on a logarithmic scale.

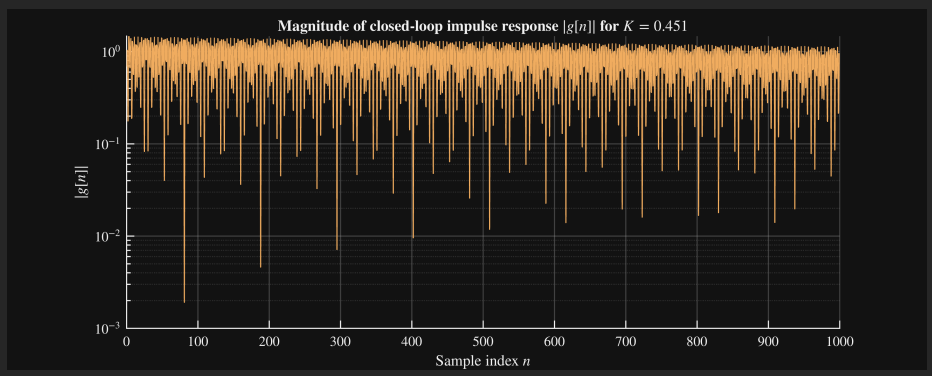

Figure 5: Magnitude of the closed-loop impulse response on a log scale.

Solution v

The oscillations clearly decay as n increases, which confirms that the closed-loop system is BIBO stable when

$$\boxed{K = 0.451.}$$

Final Answers

$$\boxed{N_{\text{ceil}} \approx 336}$$

$$\boxed{R_{\max} \approx 1.314446}$$

$$\boxed{\log_{10}(R_{\max}) \approx \frac{40}{N_{\text{ceil}}}}$$

$$\boxed{G(z) = \frac{H(z)}{1 + KH(z)} = \frac{B(z)}{A(z) + KB(z)}}$$

$$\boxed{K_{\min} = 0.451}$$

End of Project Writeup